SAGAR PUBLIC SCHOOL

Integrated Programme | Topic: Current Electricity – Level 2

PRACTICE SHEET – LEVEL 2

Current Electricity – Combination of Resistances

Time Allowed: 60 Mins **Total Questions: 20**

Name: _____ Class/Sec: _____ Date: _____

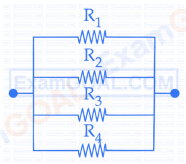
Instructions: Attempt all questions. Each question carries one correct option. Questions without options are numerical-value type: write the answer in the blank. The tag after each question shows its source examination. A self-assessment page follows; the answer key is provided as a separate sheet.

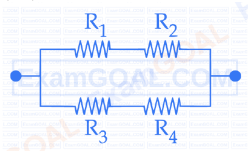
Q.1 A wire of resistance $20\,\Omega$ is divided into 10 equal parts, resulting pairs. A combination of two parts are connected in parallel and so on. Now resulting pairs of parallel combination are connected in series. The equivalent resistance of final combination is _____ Ω. *[JEE Main 2024, 5 Apr, Eve.]*

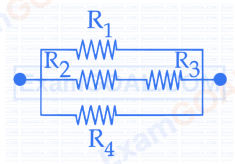
Q.2 10 resistors each of resistance $10\,\Omega$ can be connected in such as to get maximum and minimum equivalent resistance. The ratio of maximum and minimum equivalent resistance will be _____. *[JEE Main 2023, 10 Apr, Mor.]*

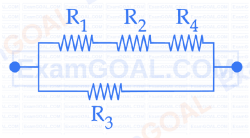
Q.3 The equivalent resistance of series combination of two resistors is s. When they are connected in parallel, the equivalent resistance is p. If $s = np$, then the minimum value for n is _____. (Round off to the Nearest Integer) *[JEE Main 2021, 17 Mar, Mor.]*

Q.4 From the combination of resistors with resistances values $R_1 = R_2 = R_3 = 5\,\Omega$ and $R_4 = 10\,\Omega$, which of the following combination is the best circuit to get an equivalent resistance of $6\,\Omega$? *[JEE Main 2025, 4 Apr, Eve.]*

(A) 

(B) 

(C) 

(D) 

Q.5 What will be the most suitable combination of three resistors $A = 2\,\Omega$, $B = 4\,\Omega$, $C = 6\,\Omega$ so that $\left(\frac{22}{3}\right)\,\Omega$ is equivalent resistance of combination? *[JEE Main 2022, 24 Jun, Eve.]*

(A) Parallel combination of A and C connected in series with B.

(B) Parallel combination of A and B connected in series with C.

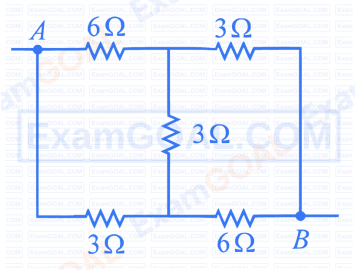
(C) Series combination of A and C connected in parallel with B.

(D) Series combination of B and C connected in parallel with A.

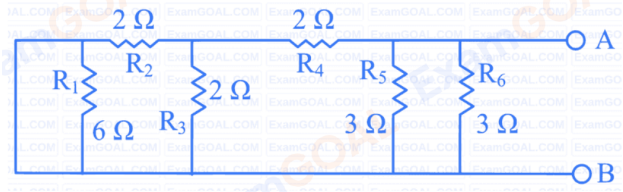
Q.6 If you are provided a set of resistances $2\,\Omega$, $4\,\Omega$, $6\,\Omega$ and $8\,\Omega$. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\,\Omega$. *[JEE Main 2021, 26 Aug, Eve.]*

(A) $4\,\Omega$ and $6\,\Omega$ are in parallel with $2\,\Omega$ and $8\,\Omega$ in series
(B) $6\,\Omega$ and $8\,\Omega$ are in parallel with $2\,\Omega$ and $4\,\Omega$ in series
(C) $2\,\Omega$ and $6\,\Omega$ are in parallel with $4\,\Omega$ and $8\,\Omega$ in series
(D) $2\,\Omega$ and $4\,\Omega$ are in parallel with $6\,\Omega$ and $8\,\Omega$ in series

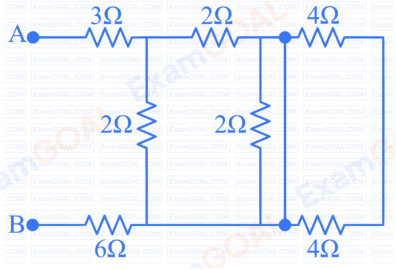
Q.7 The equivalent resistance between the points A and B in the following circuit is $\frac{x}{5}\,\Omega$. The value of x is _____. *[JEE Main 2026, 28 Jan, Mor.]*



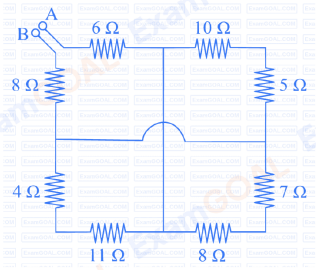
Q.8 Equivalent resistance of the following network is _____ Ω. *[JEE Main 2024, 31 Jan, Mor.]*



Q.9 In the given circuit, the equivalent resistance between the terminal A and B is _____ Ω. *[JEE Main 2023, 25 Jan, Mor.]*

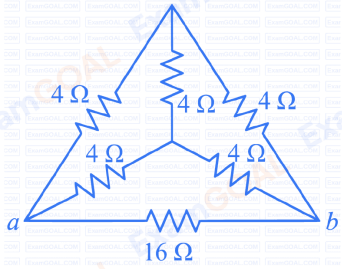


Q.10 The equivalent resistance between A and B is: *[JEE Main 2024, 9 Apr, Mor.]*



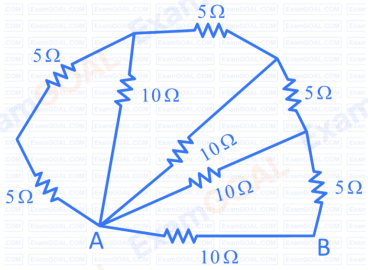
(A) $18\,\Omega$ (B) $27\,\Omega$ (C) $19\,\Omega$ (D) $25\,\Omega$

Q.11 The equivalent resistance of the circuit shown below between points a and b is: *[JEE Main 2023, 10 Apr, Mor.]*



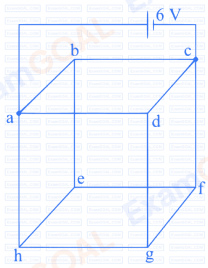
(A) $16\,\Omega$ (B) $20\,\Omega$ (C) $3.2\,\Omega$ (D) $24\,\Omega$

Q.12 The equivalent resistance between points A and B in the given network is: *[JEE Main 2022, 26 Jun, Eve.]*



(A) $65\,\Omega$ (B) $20\,\Omega$ (C) $5\,\Omega$ (D) $2\,\Omega$

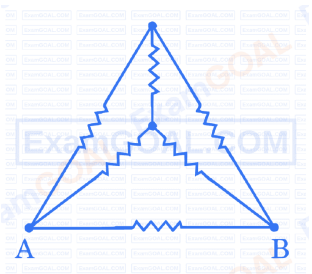
Q.13 Twelve wires each having resistance $2\,\Omega$ are joined to form a cube. A battery of 6 V emf is joined across point a and c. The voltage difference between e and f is _____ V. *[JEE Main 2024, 4 Apr, Mor.]*



Q.14 A regular hexagon is formed by six wires each of resistance $r\,\Omega$ and the corners are joined to the centre by wires of same resistance. If the current enters at one corner and leaves at the opposite corner, the equivalent resistance of the hexagon between the two opposite corners will be *[JEE Main 2026, 24 Jan, Eve.]*

(A) $\frac{3}{5}r$ (B) $\frac{4}{5}r$ (C) $\frac{5}{8}r$ (D) $\frac{3}{4}r$

Q.15 A wire of resistance R is bent into a triangular pyramid as shown in figure with each segment having same length. The resistance between points A and B is R/n. The value of n is: *[JEE Main 2025, 7 Apr, Mor.]*



(A) 16 (B) 10 (C) 12 (D) 14

Q.16 Equivalent resistance between the adjacent corners of a regular n-sided polygon of uniform wire of resistance R would be: *[JEE Main 2023, 1 Feb, Eve.]*

(A) $\frac{(n-1)R}{n^2}$ (B) $\frac{n^2R}{n-1}$
(C) $\frac{(n-1)R}{(2n-1)}$ (D) $\frac{(n-1)R}{n}$

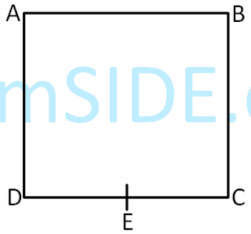
Q.17 A wire of resistance $9\,\Omega$ is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm. *[JEE Main 2025, 24 Jan, Mor.]*

Q.18 A square shaped wire with resistance of each side $3\,\Omega$ is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____. *[JEE Main 2021, 31 Aug, Mor.]*

Q.19 A metal wire of resistance $3\,\Omega$ is elongated to make a uniform wire of double its previous length. This new wire is now bent and the ends joined to make a circle. If two points on this circle make an angle $60°$ at the centre, the equivalent resistance between these two points will be: *[JEE Main 2019, 9 Apr, Eve.]*

(A) $5/2\,\Omega$ (B) $12/5\,\Omega$
(C) $7/2\,\Omega$ (D) $5/3\,\Omega$

Q.20 A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is midpoint of arm CD) *[JEE Main 2019, 9 Apr, Mor.]*



(A) $\frac{7}{64}R$ (B) $\frac{1}{16}R$
(C) R (D) $\frac{3}{4}R$

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SELF-ASSESSMENT – LEVEL 2 SHEET

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 20	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake

☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____